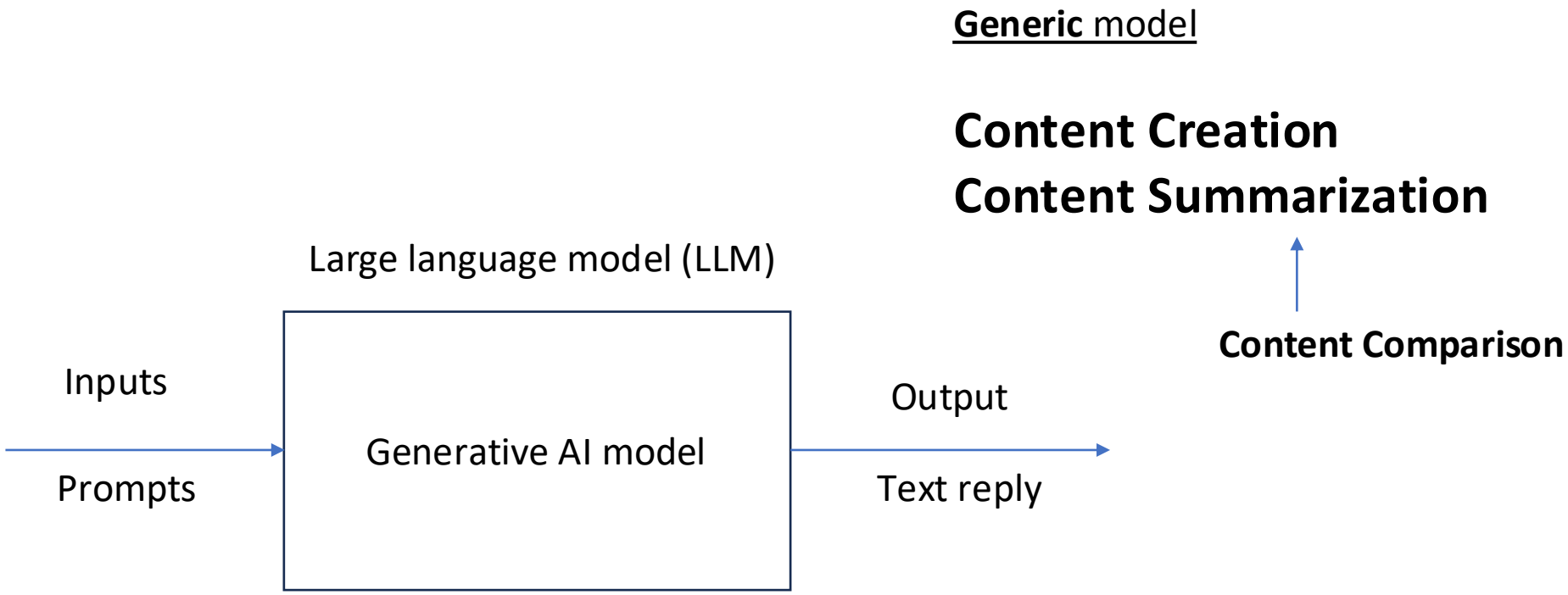
Workflow Automation with AI Solutions



Agenda

- Overview of LLM solutions
- AI Solution framework
- OCR Detection Engine
- Class Exercises
- Telegram API Setup
- Class Exercises

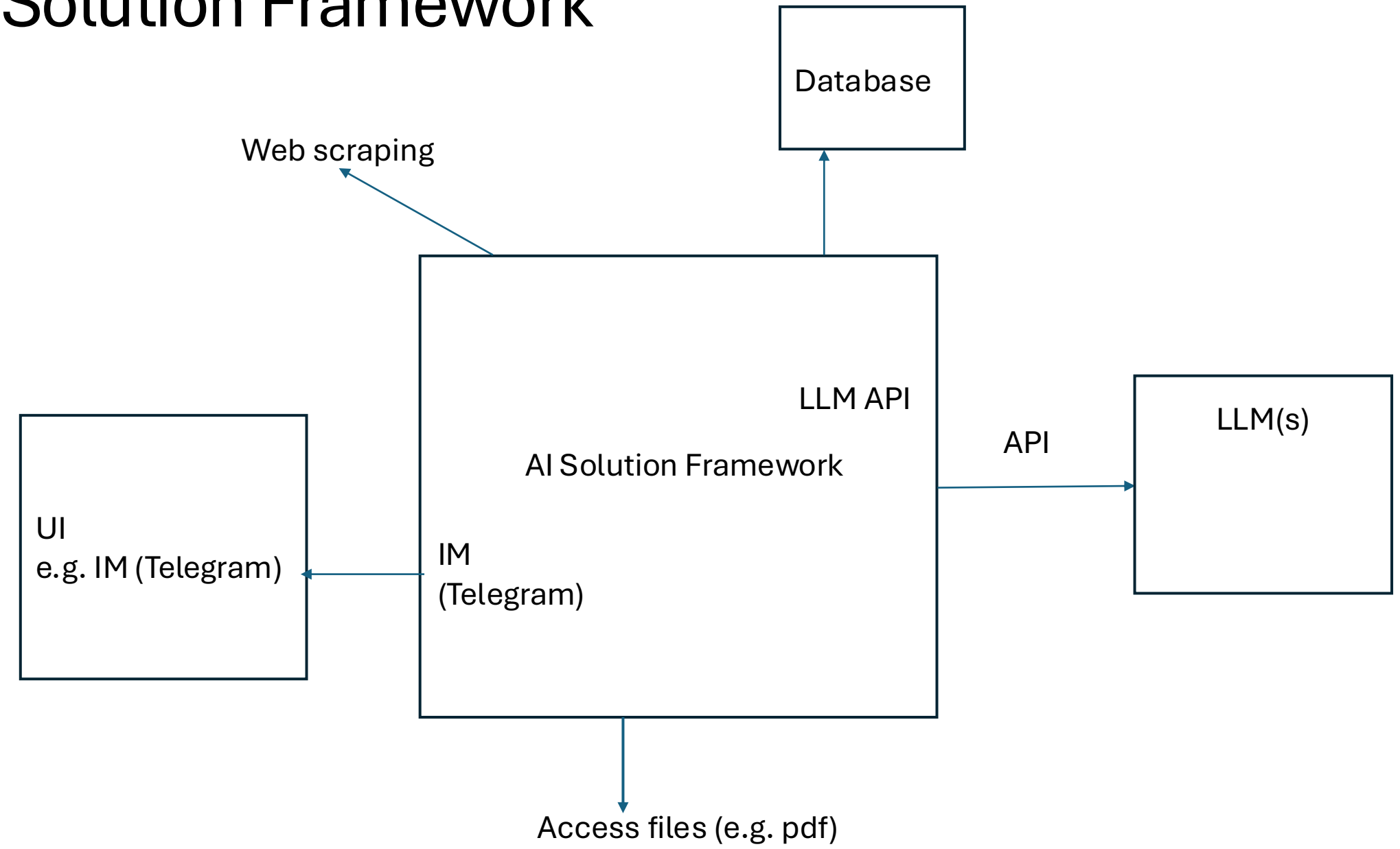
What is Generative AI



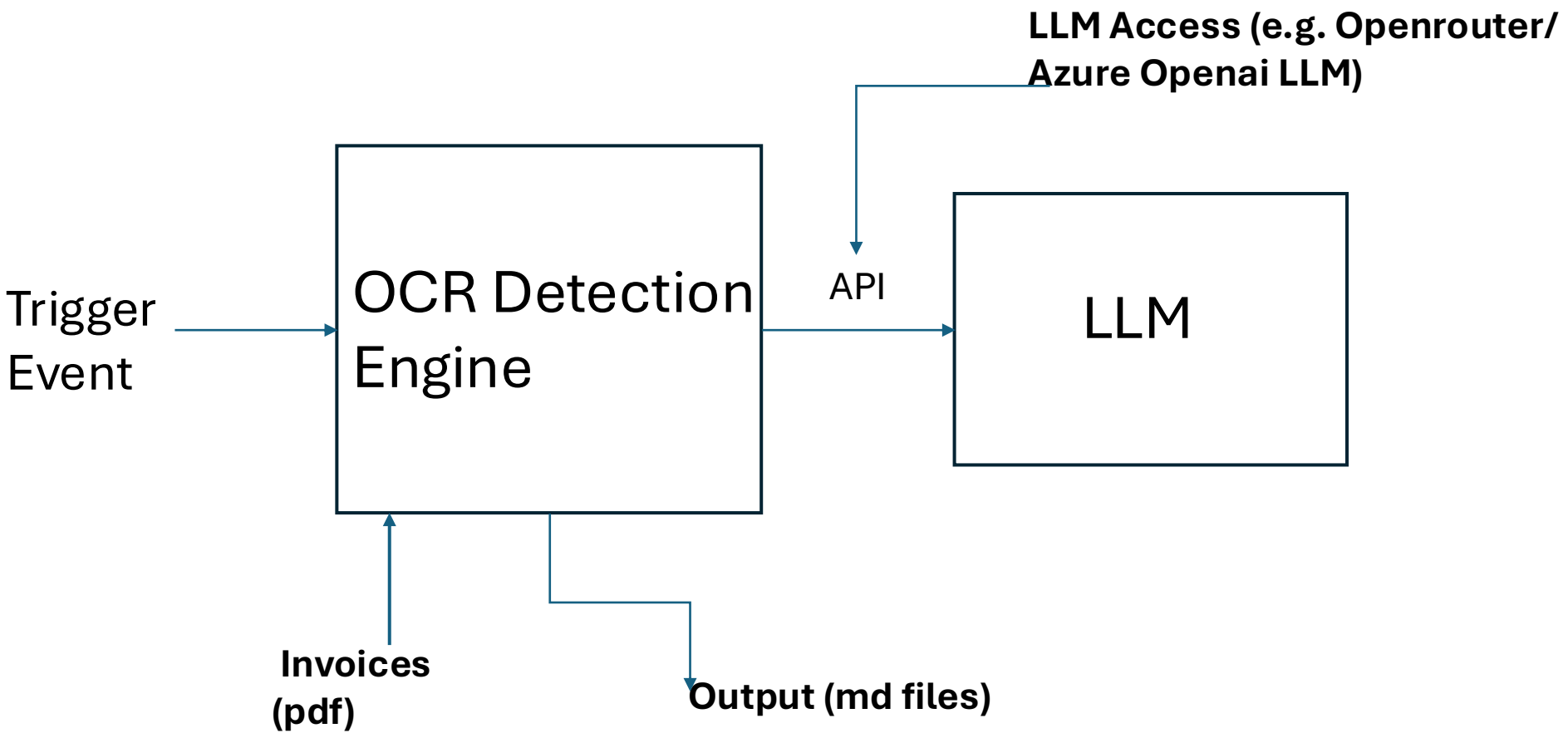
Types of LLM

What your AI agent needs to do	Best LLM type to pick	Top examples (2025 updates)	Main reason
Be super smart & handle almost any task	Flagship all-rounders	Claude 4.1 Opus, Grok 4, GPT-4o, Gemini 2.5 Pro	Highest intelligence & versatility; excels in multi-step agent workflows
Run fast, cheap, or in huge numbers	Medium open models	Llama 4 70B, Qwen3 72B	Best speed/price ratio (~$0.20/M tokens); MoE for efficient scaling
Work on phone or stay 100% private/offline	Tiny on-device models	Llama 3.2 3B, Phi-3.5 mini, Gemma 3 2B	Runs locally on <4GB RAM; no cloud dependency for privacy
Read entire books, huge codebases, long videos	Long-context beasts	Gemini 2.5 Pro (2M+ tokens), Claude 4 (1M beta)	Remembers massive inputs; 80%+ accuracy on long-doc benchmarks
See and understand images, screenshots, charts	Vision models	GPT-4o, Gemini 2.5, Qwen2.5-VL	Native multimodal (text+image/video); top on VQA/OCR tasks

AI Solution Framework



OCR Detetcion Engine



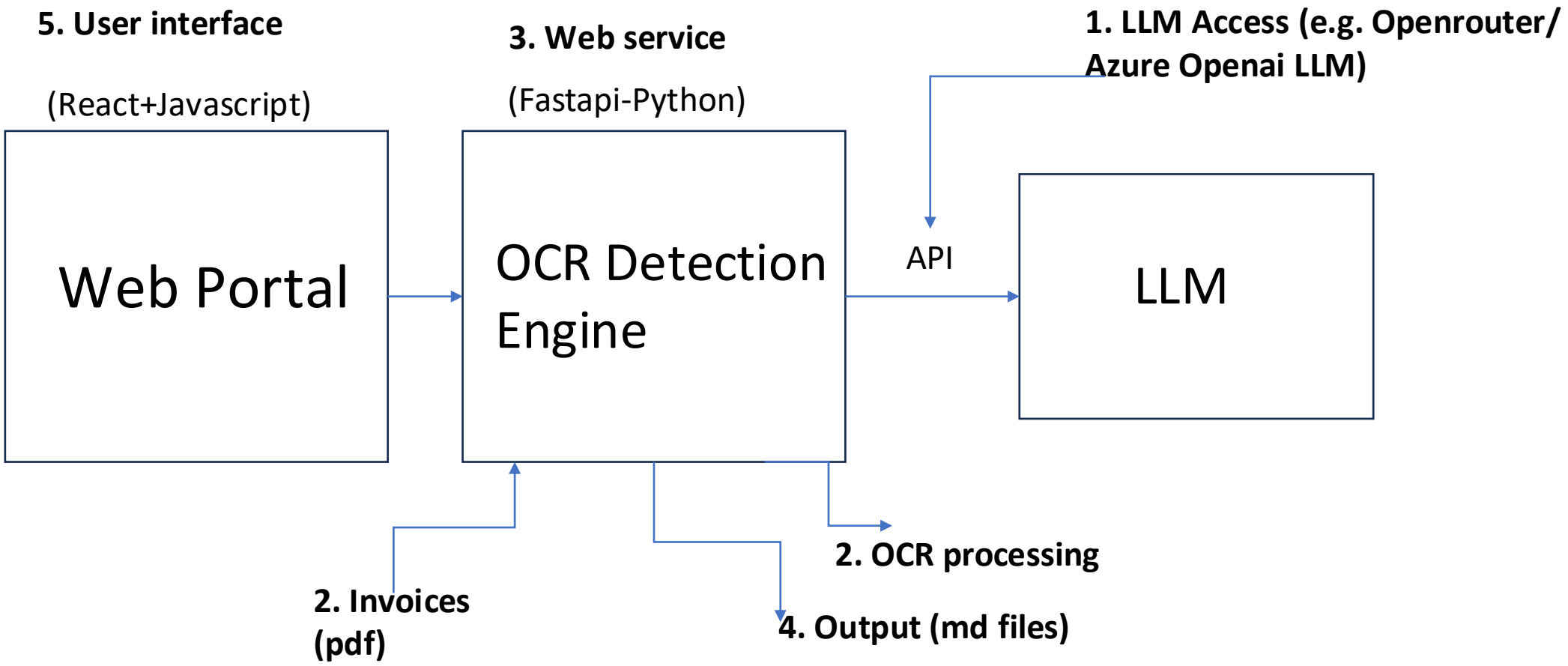
Document Processing Engine (OCR Detection Engine)

- Develop the bare minimum code snippets for the testing. (Refer to Jupyter notebook for details)
- Define the SDD (Spec driven document) for the solution
- Develop the system with IDE with coding agent supported

Class Exercise One

- In the given Jupyter notebook you could see the code snippets for OCR detection with single file.
- Use Vibe coding approach perform the following tasks:
  - Study and understand what the program logic in the code snippets
  - Modify the codes so that it could now support multiple files input
- Create the automated task which will be triggered on daily base.

Use Case: OCR Detection Engine

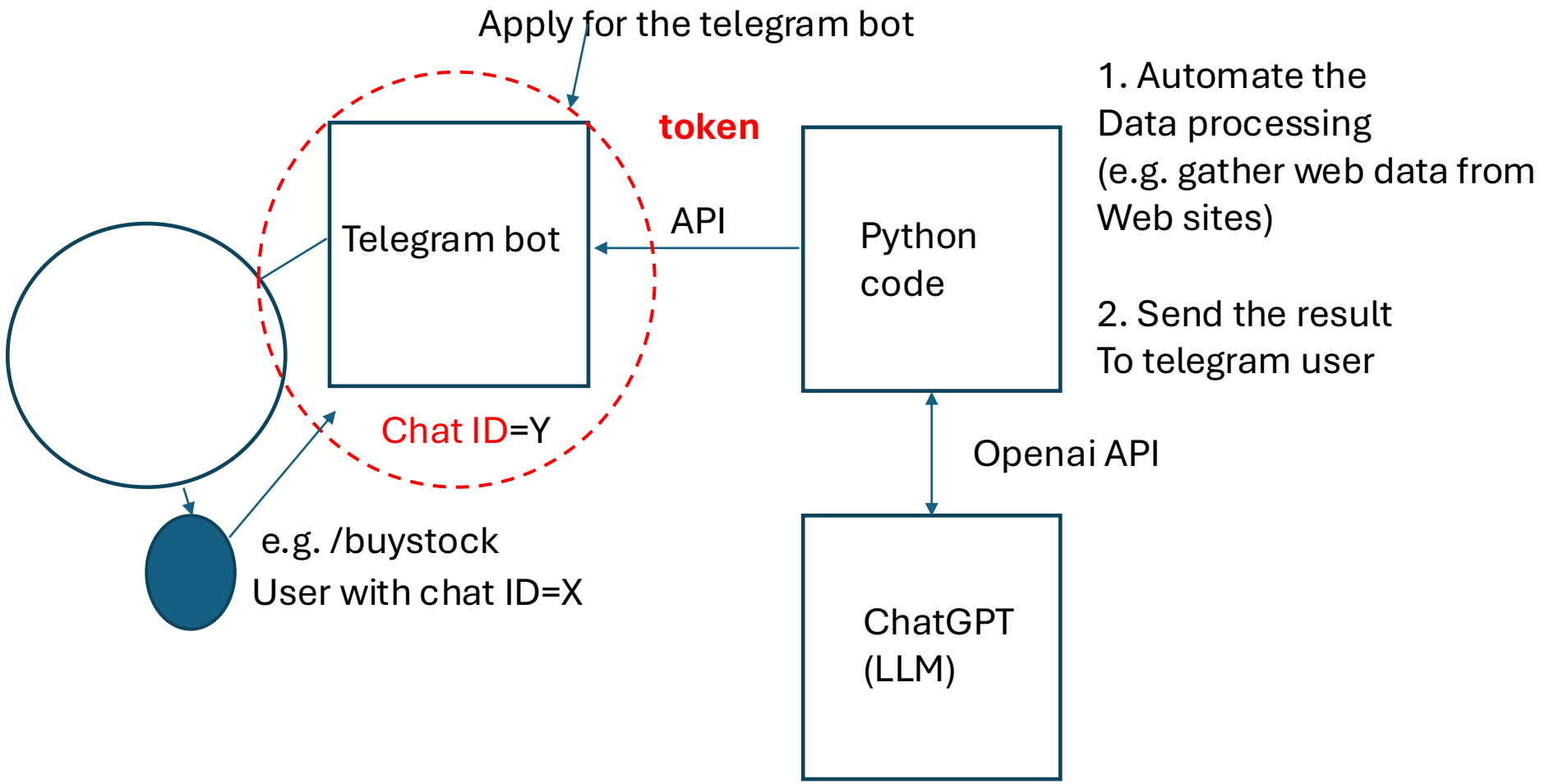


Class Exercise Two

- Develop the SDD (Spec Driven Document) based on the OCR detection engine.
- Develop the system based on the SDD file.

Telegram API

Telegram API with Python



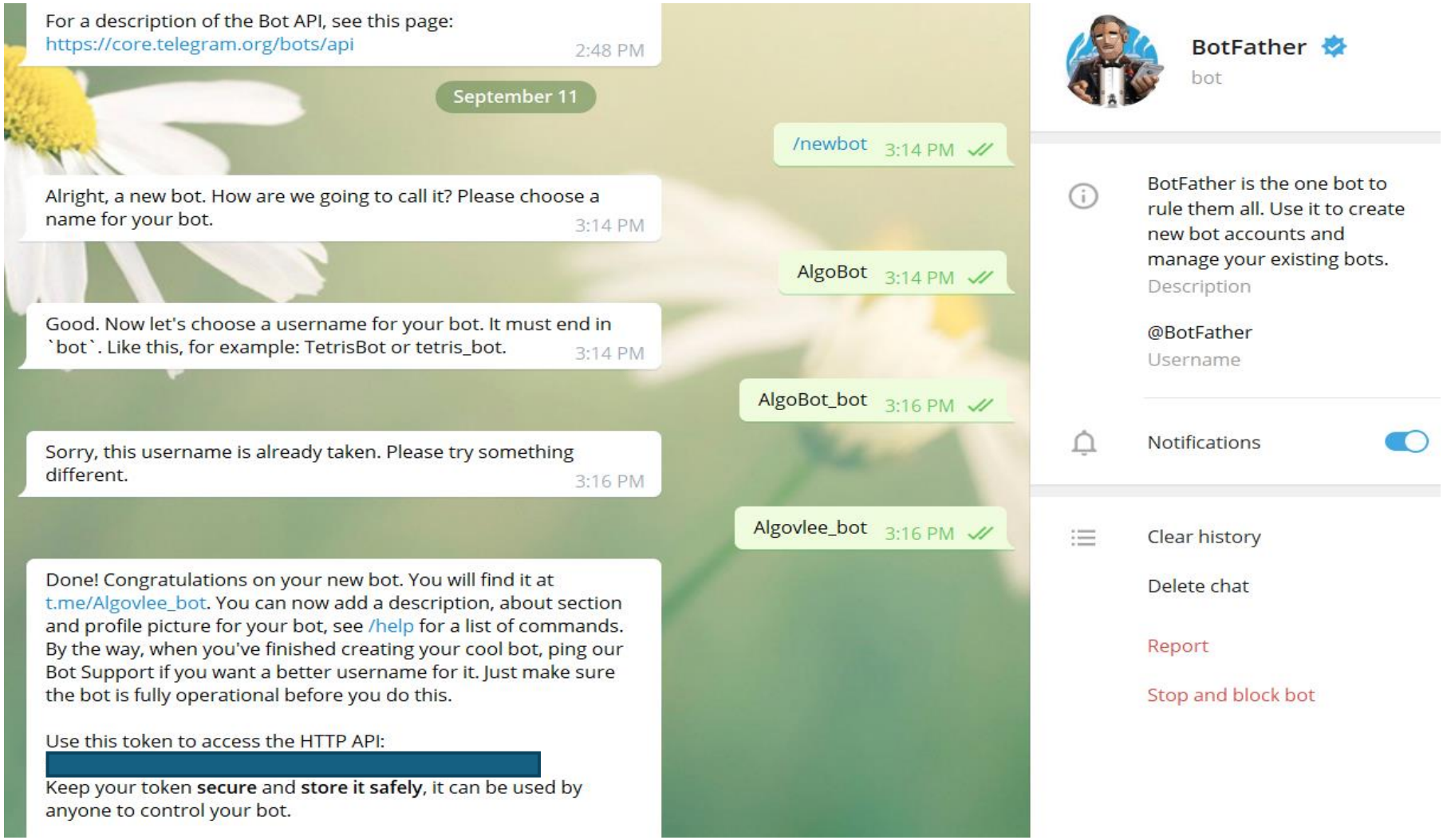
Step one

- Create Telegram account
- Login with your mobile phone
- Download Telegram desktop application

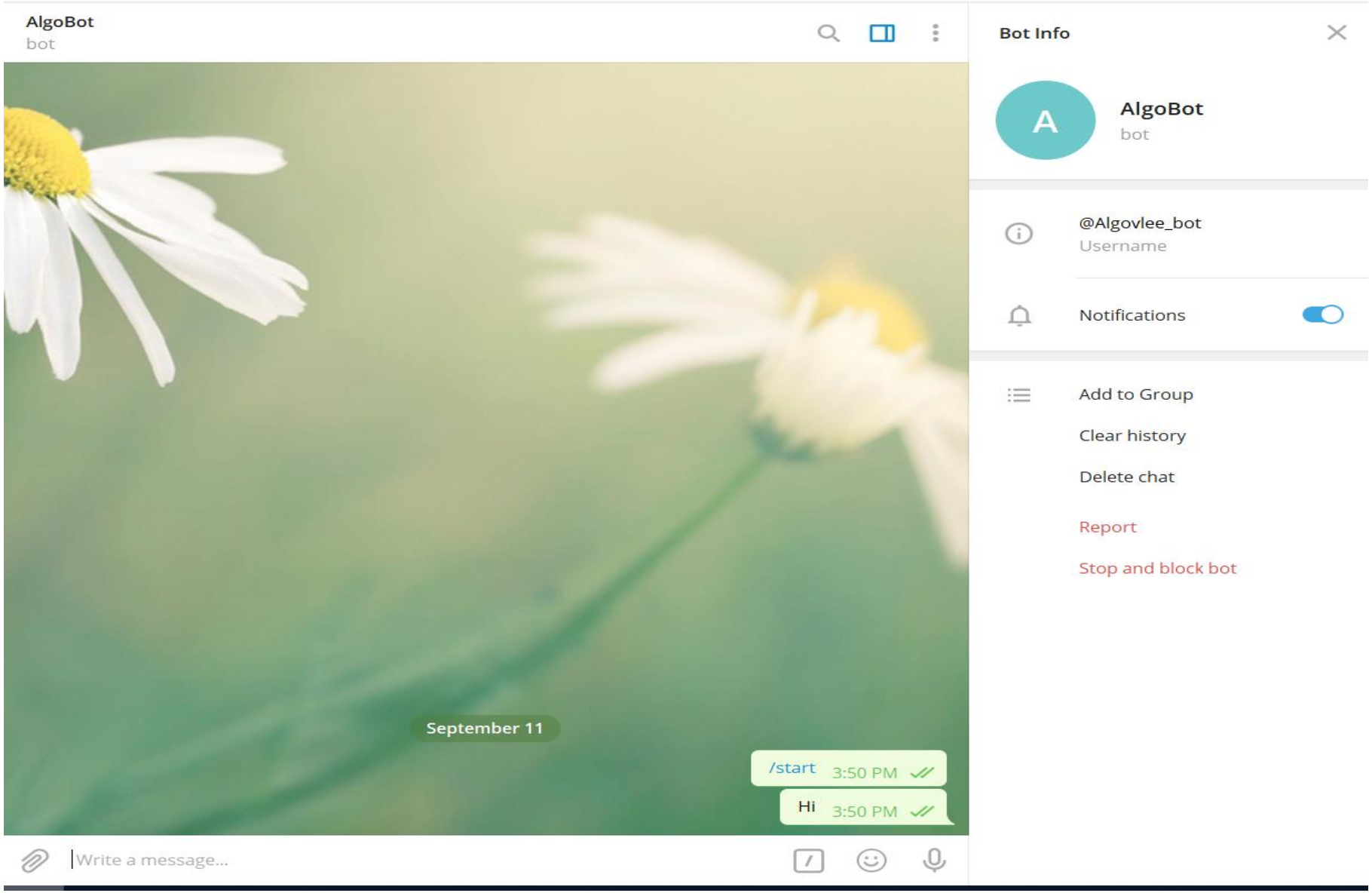
Step two

- To setup a bot in the Telegram:
- Open a chat with the **‘BotFather’** from within Telegram.
- Type in the command **/newbot**
- It will prompt you to enter a **bot name** and send you a access token.
- Open a new chat with your newly created bot.
- Click startbot
- Type something in to activate the chat.

Example


For a description of the Bot API, see this page:
https://core.telegram.org/bots/api
2:48 PM
September 11
/newbot 3:14 PM
Alright, a new bot. How are we going to call it? Please choose a name for your bot.
3:14 PM
AlgoBot 3:14 PM
Good. Now let's choose a username for your bot. It must end in `bot`. Like this, for example: TetrisBot or tetris_bot.
3:14 PM
AlgoBot_bot 3:16 PM
Sorry, this username is already taken. Please try something different.
3:16 PM
Algovlee_bot 3:16 PM
Done! Congratulations on your new bot. You will find it at t.me/Algovlee_bot. You can now add a description, about section and profile picture for your bot, see /help for a list of commands. By the way, when you've finished creating your cool bot, ping our Bot Support if you want a better username for it. Just make sure the bot is fully operational before you do this.
Use this token to access the HTTP API:
[redacted]
Keep your token secure and store it safely, it can be used by anyone to control your bot.
BotFather
bot
BotFather is the one bot to rule them all. Use it to create new bot accounts and manage your existing bots.
Description
@BotFather
Username
Notifications
Clear history
Delete chat
Report
Stop and block bot

Open a new chat with your newly created bot



Step three

- Go to the following URL – **https://api.telegram.org/botxxxxxxxxxxxxxx/getUpdates** – replacing the XXX with your access token. Here you should see a JSON structure. Note down the id (not to be confused with update_id or message_id).
- We would like to get the chat ID from here

Step three -screenshot

Your chat ID



- {"ok":true,"result":[{"update_id":XXXXXX,
- "message":{"message_id":21,"from":{"id":**XXXXX**,"is_bot":false,"first_name":"Victor Lee","username":"vickslee","language_code":"en"},"chat":{"id":XXXX,"first_name":"Victor.....

Note: You may need to enter some initial text in your chatbot and **refresh** the web link

Step four

- Go to Jupyter python
- Install package
- ! pip install python-telegram-bot
- Define your token and chat ID
- Done ☺

Class Exercise three

- Given the original code snippets from exercise one, modify it so that it could send the result (combined result) via Telegram.

Use Case: OCR Detection Engine

